Transformation in ETL

Question 1 : Define Data Transformation in ETL and explain why it is important.

Answer :- Data transformation is the process used to convert the data into relevant structure, so that it can be analyzed easily. Because in real life working the raw data contains null values, duplicates, wrong data type etc. So it becomes a necessity to clean that data.

Question 2 : List any four common activities involved in DataCleaning.

Answer :- These are the most common data cleaning activities.

1. Cleaning the duplicate values.
2. Managing null values.
3. Handling outliers.
4. Fixing spelling mistakes.

Question 3 : What is the difference between Normalization and Standardization?

Answer:-

<u>Normalization</u> is used to rescale the data into a fixed range.

<u>Standardization</u> is used to rescale the data into 0 mean and 1 standard deviation.

Question 4 : A dataset has missing values in the "Age" column. Suggest two techniques to handle this and explain when they should be used.

Answer :-

Use Mean or median of age column

Or

Delete row if the less values are missing.

Or

Delete the complete column if more than 70% rows contain missing value in the age column.

Question 5 : Convert the following inconsistent "Gender" entries into a standardized format ("Male", "Female"): ["M", "male", "F", "Female", "MALE", "f"]

Answer :-

Male: "M", "male", "MALE"

Female: "F", "Female", "f"

Question 6 : What is One-Hot Encoding? Give an example with the categories: “Red, Blue, Green”.

Answer :- It converts categorical variables into binary (0 or 1) columns.

Red: [1, 0, 0]

Blue: [0, 1, 0]

Green: [0, 0, 1]

Question 7 : Explain the difference between Data Integration and Data Mapping in ETL.

Answer :- Data Integration:-The overall process of combining data from different sources into a single destination view.

Data Mapping:- The specific step of defining how individual data fields from one system connect to fields in another system.

Question 8 : Explain why Z-score Standardization is preferred over Min-Max Scaling when outliers exist.

Answer :- Z-score is preferred because Min-Max Scaling squashes most data into a very small range if extreme outliers exist, whereas Z-score maintains the distribution by using the mean = 0 and standard deviation =1.